Cryptocurrency mining (Monero)

Presented by Finnegan Thomsen

What is crypto mining? And how does it work?

Crypto mining is working within a blockchain, to process and finalize transactions. The miners actually solve or process the transactions by solving a 64 bit hexadecimal number or a hash. Which is where the term “Hashrate” comes from, which will be mentioned later. Miners are rewarded with a portion of cryptocurrency once the work is done and a block is added to the blockchain.

*A blockchain is a record of all transactions that involve a certain cryptocurrency and all users that have interacted with it.

Creating your own miner.

The specific mining software we will be using is called "xmrig".

I have created a github repository for easy installation of the mining software.

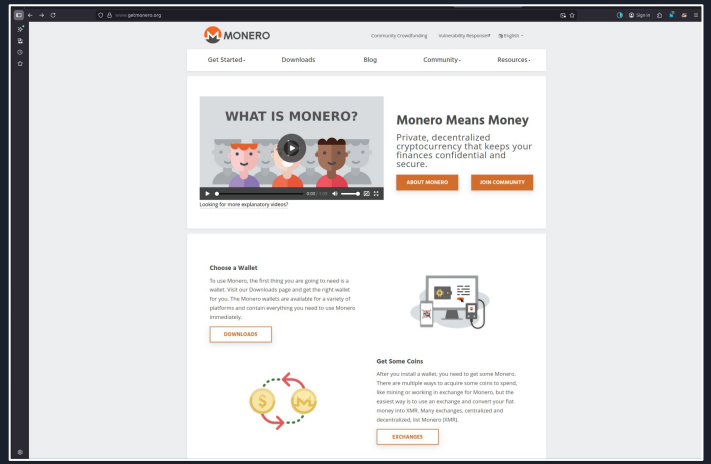
Git clone "https://github.com/Finnegannnn/xmrig_installation.git" or go to "**https://tinyurl.com/bdfazkxc**"

Once you have the repository cloned, cd into it. Then modify the permissions of the file "Install.sh" to a+x (sudo chmod a+x install.sh). Then run the file using ./install.sh.

Once the script is finished it will list where the mining software build is located, don't navigate there yet. Pause. We now have to create a wallet.

*Xmrig

Creating your own crypto wallet

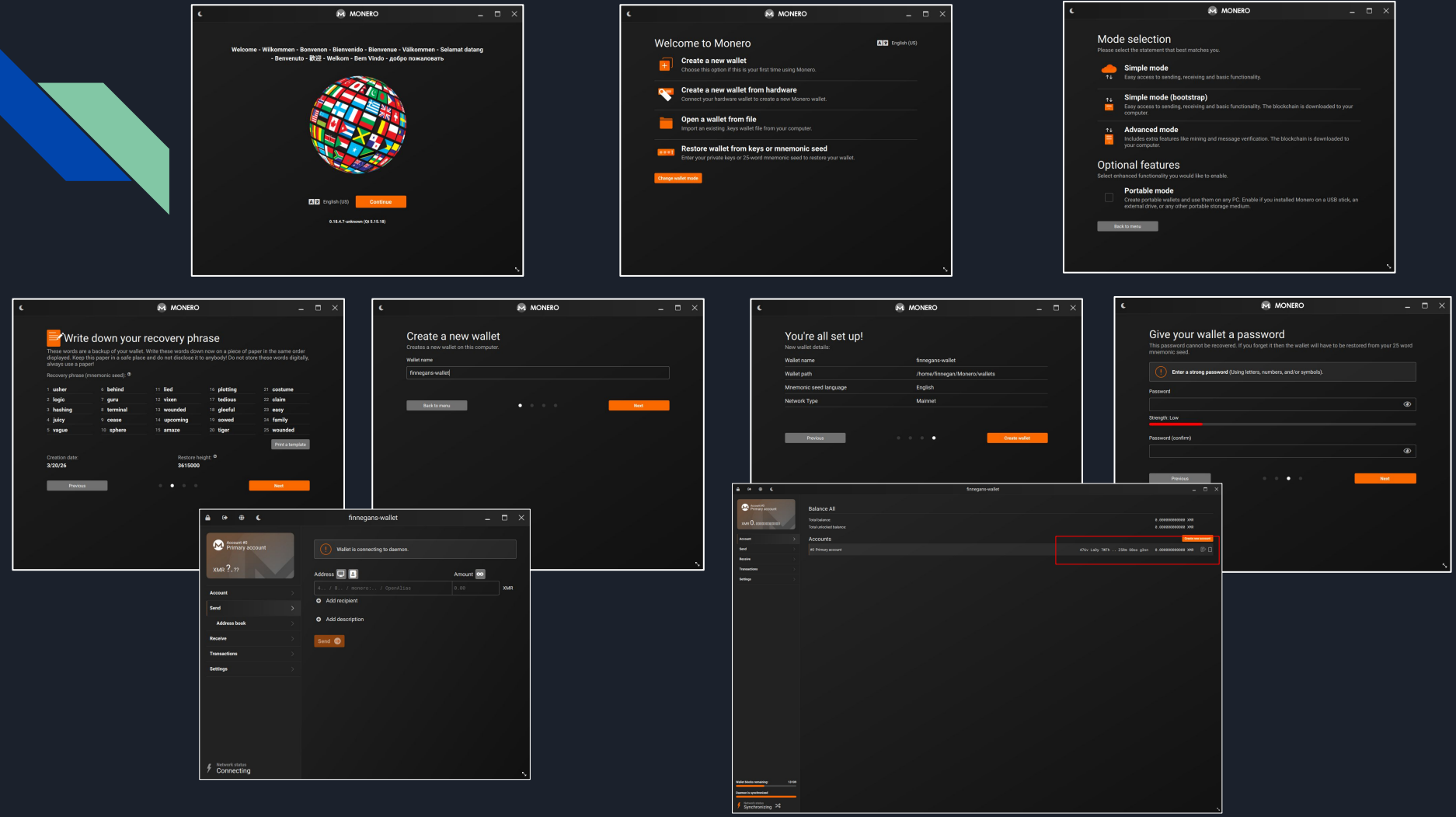


Visit “getmonero.com”, and then navigate downloads. You’ll then download the Linux 64 bit edition.

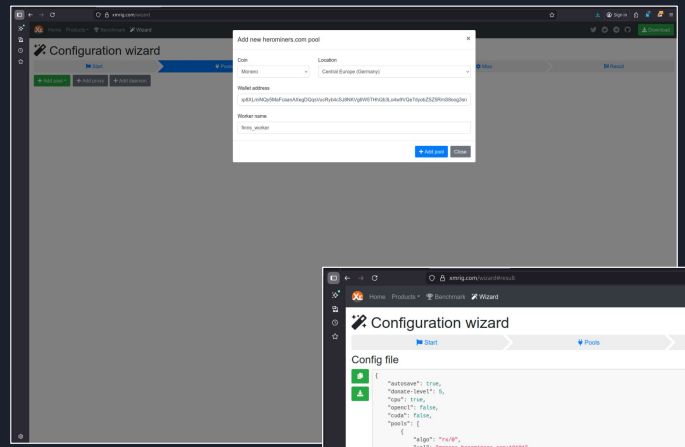
Once the file is downloaded, you’ll have to extract it using “tar”, using the command “tar -xf filename.tar.bz2”.

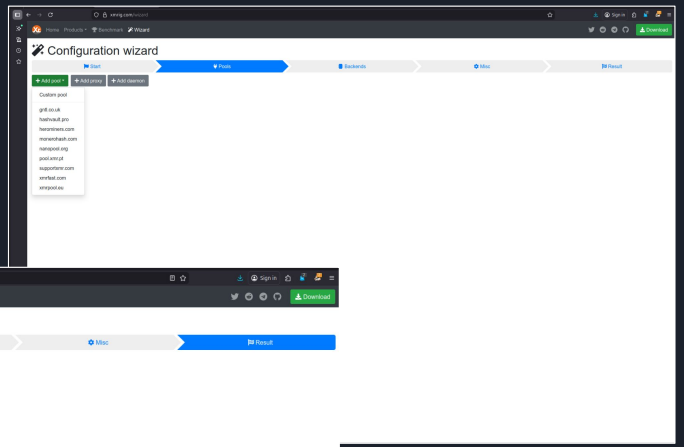
Then navigate into the newly created directory, and run “./monero-wallet-gui”.

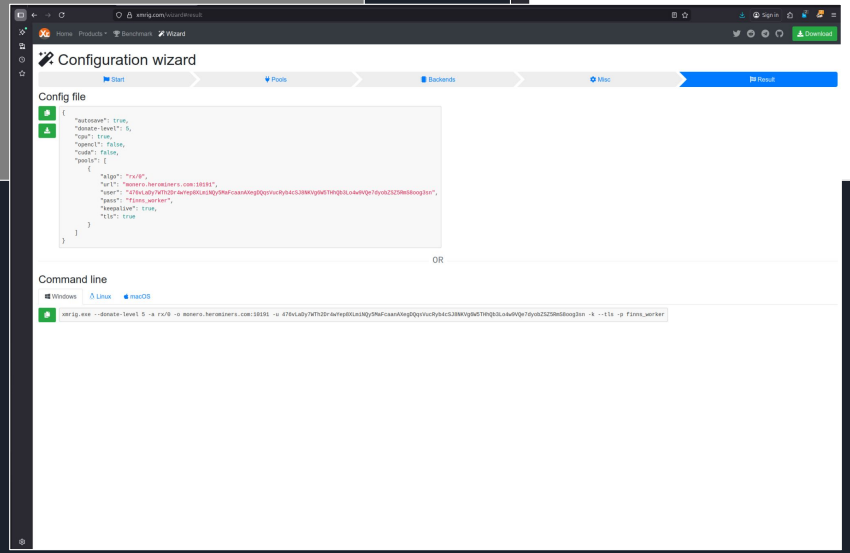
Once your in the software begin creating a wallet, steps will be shown on the next slide.


MONERO
Welcome - Willkommen - Bonvenon - Bienvenido - Bienvenue - Välkommen - Selamat datang - Benvenuto - 欢迎 - Welkom - Bem Vindo - Добро пожаловать
English (US)
Continue
0.18.4.7-unknown (Qt 5.15.18)
Welcome to Monero
English (US)
Create a new wallet
Choose this option if this is your first time using Monero.
Create a new wallet from hardware
Connect your hardware wallet to create a new Monero wallet.
Open a wallet from file
Import an existing .keys wallet file from your computer.
Restore wallet from keys or mnemonic seed
Enter your private keys or 25-word mnemonic seed to restore your wallet.
Change wallet mode
Mode selection
Please select the statement that best matches you.
Simple mode
Easy access to sending, receiving and basic functionality.
Simple mode (bootstrap)
Easy access to sending, receiving and basic functionality. The blockchain is downloaded to your computer.
Advanced mode
Includes extra features like mining and message verification. The blockchain is downloaded to your computer.
Optional features
Select enhanced functionality you would like to enable.
Portable mode
Create portable wallets and use them on any PC. Enable if you installed Monero on a USB stick, an external drive, or any other portable storage medium.
Back to menu
Write down your recovery phrase
These words are a backup of your wallet. Write these words down now on a piece of paper in the same order displayed. Keep this paper in a safe place and do not disclose it to anybody! Do not store these words digitally, always use a paper!
Recovery phrase (mnemonic seed)
1 usher 6 behind 11 lied 16 plotting 21 costume
2 logic 7 guru 12 vixen 17 tedious 22 claim
3 hashing 8 terminal 13 wounded 18 gleeful 23 easy
4 juicy 9 cease 14 upcoming 19 sowed 24 family
5 vague 10 sphere 15 amaze 20 tiger 25 wounded
Print a template
Creation date:
3/20/26
Restore height:
3615000
Previous
Next
Create a new wallet
Creates a new wallet on this computer.
Wallet name
finnegans-wallet
Back to menu
Next
You're all set up!
New wallet details:
Wallet name finnegans-wallet
Wallet path /home/finnegan/Monero/wallets
Mnemonic seed language English
Network Type Mainnet
Previous
Create wallet
Give your wallet a password
This password cannot be recovered. If you forget it then the wallet will have to be restored from your 25 word mnemonic seed.
Enter a strong password (Using letters, numbers, and/or symbols).
Password
Strength: Low
Password (confirm)
Previous
Next
finnegans-wallet
Account #0
Primary account
XMR ?.??
Account
Send
Address book
Receive
Transactions
Settings
Wallet is connecting to daemon.
Address
Amount
4.. / 8.. / monero:.. / OpenAlias
0.00
XMR
Add recipient
Add description
Send
Network status
Connecting
Balance All
Total balance
Total unlocked balance
Accounts
Create new account
#0 Primary account
Wallet blocks remaining:
Daemon is synchronized
Synchronizing

Creating your own miner configuration file







Select new configuration, then pool, and then "Monerohash.com". From there, then paste your wallet address

Continue through the steps and then once the final step is reached copy the linux version of the command line command

```
/home/finnegan/.zshrc:8: command not found: #
~/cryto_mining/xmrig/build | master                                                                    14:33:26
> ls
CMakeCache.txt  CMakeFiles  cmake_install.cmake  config.json  libxmrig-asm.a  Makefile  src  xmrig

~/cryto_mining/xmrig/build | master                                                                    14:33:30
> torsocks ./xmrig --donate-level 5 -o monerohash.com:9999 -u 49qYKFaV4P6esBFa7tdFe8cHiL72EcaqieWajTtVWiGiNJk8Q4DSsZBVGGGx89LyaSNMFD5P8DJDS3q8hwfqJzEGCTpUPJY -k --tls
```

Then paste the command, making sure to include “torsocks” at the start, and voila.

```
~/cryto_mining/xmrig/build | master                                                                    14:30:06
> torsocks ./xmrig --donate-level 5 -o monerohash.com:9999 -u 49qYKFaV4P6esBFa7tdFe8cHiL72EcaqieWajTtVWiGiNJk8Q4DSsZBVGGGx89LyaSNMFD5P8DJDS3q8hwfqJzEGCTpUPJY -k --tls
 * ABOUT        XMRig/6.25.0 gcc/13.3.0 (built for Linux x86-64, 64 bit)
 * LIBS         libuv/1.48.0 OpenSSL/3.0.13 hwloc/2.10.0
 * HUGE PAGES   supported
 * 1GB PAGES    disabled
 * CPU          13th Gen Intel(R) Core(TM) i7-1360P (1) 64-bit AES
                    9.0 MB    18.0 MB 12C 16T      1
 * MEMORY       7.8/29.1 GB
 * DONATE       5%
 * ASSEMBLY     auto:intel
 * POOL #1      monerohash.com:9999 algo auto
 * COMMANDS     hashrate, pause, resume, results, connection
 * OPENCL       disabled
 * CUDA         disabled
[2026-04-10 14:30:13    ] net      use pool monerohash.com:9999 TLSv1.2
[2026-04-10 14:30:13    ] net
[2026-04-10 14:30:13    ] net      new job from monerohash.com:9999 diff 150000 algo rx/0 height 3649646 (15 tx)
[2026-04-10 14:30:13    ] cpu      use argon2 implementation AVX2
[2026-04-10 14:30:13    ] msr      cannot read MSR 0x000001a4
[2026-04-10 14:30:13    ] msr      FAILED TO APPLY MSR MOD, HASHRATE WILL BE LOW
[2026-04-10 14:30:13    ] randomx  init dataset algo rx/0 (16 threads)
[2026-04-10 14:30:13    ] randomx  allocated 2336 MB              huge pages 0% 0/1168 +JIT
[2026-04-10 14:30:17    ] randomx  dataset ready
[2026-04-10 14:30:17    ] cpu      use profile  rx  (13 threads) scratchpad 2048 KB
[2026-04-10 14:30:17    ] cpu      READY threads 13/13 (13) huge pages 0% 0/13 memory 26624 KB
[2026-04-10 14:30:32    ] net      new job from monerohash.com:9999 diff 150000 algo rx/0 height 3649646 (21 tx)
[2026-04-10 14:30:52    ] net      new job from monerohash.com:9999 diff 150000 algo rx/0 height 3649646 (27 tx)
[2026-04-10 14:31:10    ] cpu      accepted (1/0) diff 150000
[2026-04-10 14:31:12    ] net      new job from monerohash.com:9999 diff 150000 algo rx/0 height 3649646 (31 tx)
[2026-04-10 14:31:16    ] cpu      accepted (2/0) diff 150000
[2026-04-10 14:31:17    ] cpu      accepted (3/0) diff 150000
[2026-04-10 14:31:17    ] miner    speed 10s/60s/15m 3071.2 n/a n/a H/s max 3161.1 H/s
[2026-04-10 14:31:18    ] net      new job from monerohash.com:9999 diff 225008 algo rx/0 height 3649646 (31 tx)
[2026-04-10 14:31:21    ] cpu      accepted (4/0) diff 225008
[2026-04-10 14:31:32    ] net      new job from monerohash.com:9999 diff 225008 algo rx/0 height 3649646 (34 tx)
[2026-04-10 14:31:43    ] net      new job from monerohash.com:9999 diff 225008 algo rx/0 height 3649647 (21 tx)
[2026-04-10 14:31:48    ] net      new job from monerohash.com:9999 diff 337521 algo rx/0 height 3649647 (21 tx)
[2026-04-10 14:32:03    ] net      new job from monerohash.com:9999 diff 337521 algo rx/0 height 3649647 (23 tx)
[2026-04-10 14:32:17    ] miner    speed 10s/60s/15m 2806.2 2932.2 n/a H/s max 3161.1 H/s
```